CISCO PACKET TRACER 9.0

University-Level Network Project Guide

Router & Switch Management

with TFTP Backup and IOS Upgrade

Complete Step-by-Step Implementation Guide with Full CLI Commands

Project Highlights

15 Devices | 4 Departments | VLAN Segmentation | OSPF Routing | TFTP Backup

Designed for Cisco Packet Tracer 9.0

Table of Contents

1. Project Overview & Objectives 4
   1.1 Project Objectives 4
   1.2 Learning Outcomes 4
2. Network Topology Design 5
   2.1 Topology Overview 5
   2.2 VLAN Design 5
   2.3 IP Addressing Scheme 6
3. Cisco Packet Tracer Setup 7
   3.1 Software Requirements 7
   3.2 Device Selection in Packet Tracer 7
   3.3 Physical Topology Connections 7
4. End Device IP Configuration 9
   4.1 TFTP Server Configuration 9
   4.2 DNS / Web Server Configuration 9
   4.3 Management PC 9
   4.4 Department PCs 10
5. Switch Configuration 11
   5.1 Distribution Switch 1 (Dist-SW-1) — L3 Switch 11
   5.2 Distribution Switch 2 (Dist-SW-2) — L3 Switch 15
   5.3 Access Switch 1 (Access-SW-1) — Admin Department 17
   5.4 Access Switch 2 (Access-SW-2) — IT Department 18
   5.5 Access Switch 3 (Access-SW-3) — Computer Labs 19
   5.6 Access Switch 4 (Access-SW-4) — Library 20
6. Router Configuration 22
   6.1 Core Router (CR-1) 22
   6.2 Edge Router (ER-1) 23
7. OSPF Verification & Connectivity Testing 25
   7.1 Verify OSPF Neighbor Relationships 25
   7.2 End-to-End Connectivity Tests 25
8. TFTP Configuration Backup 26
   8.1 How TFTP Works in CPT 26
   8.2 Enable TFTP Service on Server 26
   8.3 Backup Running Configuration from CR-1 26
   8.4 Backup Startup Configuration from CR-1 26
   8.5 Backup All Devices — Command Reference 27

8.6 Verify Backup on TFTP Server 27
8.7 Restore Configuration from TFTP 27
9. IOS Image Backup and Upgrade via TFTP 29
9.1 Understanding IOS Image Management 29
9.2 Step 1 — Check Current IOS Version 29
9.3 Step 2 — Back Up Current IOS to TFTP 29
9.4 Step 3 — Simulate Loading New IOS from TFTP 30
9.5 Step 4 — Configure Boot System for New IOS 30
10. Advanced Configuration & Security 32
10.1 Access Control Lists (ACLs) 32
10.2 Spanning Tree Protocol (STP) Optimization 32
10.3 DHCP Configuration on ER-1 32
11. Complete Verification Command Reference 34
11.1 Switch Verification 34
11.2 Router Verification 34
11.3 TFTP Verification 35
12. Using Simulation Mode in Packet Tracer 36
12.1 Setting Up Simulation 36
12.2 Simulating TFTP Backup 36
12.3 Simulating OSPF Convergence 36
12.4 Verifying Inter-VLAN Traffic Path 36
13. Troubleshooting Guide 37
13.1 Common Issues & Fixes 37
13.2 Debug Commands 37
14. Project Summary & Expected Results 38
14.1 Expected Final State 38
14.2 Project File Naming Convention 38
14.3 Documentation Checklist 38
Appendix A: Quick Command Reference 40
Appendix B: Device Management IPs 41

1. Project Overview & Objectives

This project simulates an enterprise-grade university network using Cisco Packet Tracer 9.0. The topology models a multi-department university campus with centralized network management, TFTP-based configuration backup and restoration, and IOS image management.

1.1 Project Objectives

- Design and implement a hierarchical three-tier campus network topology
- Configure inter-VLAN routing using Layer 3 switching and router-on-a-stick
- Deploy OSPF as the dynamic routing protocol across all routers
- Set up a centralized TFTP server for configuration and IOS image management
- Automate configuration backup from all routers and switches to TFTP
- Simulate an IOS upgrade process via TFTP on selected devices
- Implement access control and security best practices (SSH, ACLs, port security)
- Validate full end-to-end connectivity between all VLANs and departments

1.2 Learning Outcomes

- Understand centralized network management concepts
- Master TFTP operations (copy, backup, restore) in Cisco IOS
- Practice enterprise VLAN design and trunking
- Gain hands-on experience with OSPF configuration and verification
- Learn IOS image management workflows

> **IMPORTANT:**
> All CLI commands in this guide are fully compatible with Cisco Packet Tracer 9.0. Commands were verified against CPT 9.0 IOS versions. Some advanced IOS features available on real hardware may differ slightly in CPT simulation mode.

2. Network Topology Design

2.1 Topology Overview

The network follows a three-tier hierarchical model: Core, Distribution, and Access layers. This mirrors real-world enterprise and university campus deployments.

Layer	Device Name	Role
Core Layer	Core-Router (CR-1)	Central routing hub, OSPF ABR, gateway to ISP/Internet
Distribution	Dist-SW-1 (L3 Switch)	VLAN routing for Admin & IT blocks
Distribution	Dist-SW-2 (L3 Switch)	VLAN routing for Labs & Library
Access Layer	Access-SW-1	Admin Department end-user switch
Access Layer	Access-SW-2	IT Department end-user switch
Access Layer	Access-SW-3	Computer Lab switch
Access Layer	Access-SW-4	Library switch
WAN/Edge	Edge-Router (ER-1)	WAN edge, NAT, DHCP server
Management	TFTP-Server	Configuration & IOS backup server
Management	Mgmt-PC	Network admin workstation
End Devices	Admin-PC1, Admin-PC2	Admin department hosts
End Devices	IT-PC1, IT-PC2	IT department hosts
End Devices	Lab-PC1, Lab-PC2	Computer lab hosts
End Devices	Lib-PC1	Library host
Services	DNS/Web Server	Internal services server

2.2 VLAN Design

VLANs provide logical segmentation between departments, improving security and broadcast control.

VLAN ID	Name	Subnet	Description
VLAN 10	Admin	192.168.10.0/24	Administrative Department
VLAN 20	IT	192.168.20.0/24	IT Department
VLAN 30	Labs	192.168.30.0/24	Computer Laboratories
VLAN 40	Library	192.168.40.0/24	Library Network
VLAN 50	Management	192.168.50.0/24	Network Management (TFTP, Admin PC)

VLAN 99	Native/Trunk	N/A	Native VLAN for trunk links
VLAN 100	Servers	192.168.100.0/24	Internal Servers (DNS, Web, TFTP)

2.3 IP Addressing Scheme

Device	Interface	IP Address	Notes
Core-Router (CR-1)	Fa0/0 (to Dist-SW-1)	10.0.1.1/30	Point-to-point link
Core-Router (CR-1)	Fa0/1 (to Dist-SW-2)	10.0.2.1/30	Point-to-point link
Core-Router (CR-1)	Se0/0/0 (to ER-1)	10.0.0.1/30	WAN serial link
Edge-Router (ER-1)	Se0/0/0 (to CR-1)	10.0.0.2/30	WAN serial link
Edge-Router (ER-1)	Fa0/0 (Internet-side)	DHCP/Static ISP	Internet-facing interface
Dist-SW-1	VLAN 10 SVI	192.168.10.1/24	Admin gateway
Dist-SW-1	VLAN 20 SVI	192.168.20.1/24	IT gateway
Dist-SW-1	VLAN 50 SVI	192.168.50.1/24	Mgmt gateway
Dist-SW-1	Uplink to CR-1	10.0.1.2/30	L3 routed link
Dist-SW-2	VLAN 30 SVI	192.168.30.1/24	Labs gateway
Dist-SW-2	VLAN 40 SVI	192.168.40.1/24	Library gateway
Dist-SW-2	VLAN 100 SVI	192.168.100.1/24	Servers gateway
Dist-SW-2	Uplink to CR-1	10.0.2.2/30	L3 routed link
TFTP-Server	NIC	192.168.100.10/24	GW: 192.168.100.1
Mgmt-PC	NIC	192.168.50.10/24	GW: 192.168.50.1
Admin-PC1	NIC	192.168.10.10/24	GW: 192.168.10.1
Admin-PC2	NIC	192.168.10.11/24	GW: 192.168.10.1
IT-PC1	NIC	192.168.20.10/24	GW: 192.168.20.1
IT-PC2	NIC	192.168.20.11/24	GW: 192.168.20.1
Lab-PC1	NIC	192.168.30.10/24	GW: 192.168.30.1
Lab-PC2	NIC	192.168.30.11/24	GW: 192.168.30.1
Lib-PC1	NIC	192.168.40.10/24	GW: 192.168.40.1
DNS/Web Server	NIC	192.168.100.20/24	GW: 192.168.100.1

3. Cisco Packet Tracer Setup

3.1 Software Requirements

- Cisco Packet Tracer Version: 9.0 or higher
- Operating System: Windows 10/11, macOS 12+, or Ubuntu 20.04+
- RAM: Minimum 8 GB (16 GB recommended for 15+ devices)
- Disk: 5 GB free space

3.2 Device Selection in Packet Tracer

Open Cisco Packet Tracer. Use the following device categories from the bottom toolbar:

Device	CPT Category / Model
Core-Router (CR-1)	Routers > 2911 (or 1941)
Edge-Router (ER-1)	Routers > 2911
Dist-SW-1, Dist-SW-2	Switches > 3560-24PS (multilayer)
Access-SW-1 to SW-4	Switches > 2960-24TT
TFTP-Server, DNS/Web Server	End Devices > Server
Mgmt-PC, Admin-PCs, IT-PCs, Lab-PCs, Lib-PC	End Devices > PC

> **TIP:**
> For the 3560 multilayer switch to perform L3 routing in CPT, you must manually enable the SDM template and 'ip routing' command (covered in Section 5).

3.3 Physical Topology Connections

Connect devices using the following cable types. In CPT, select the correct cable from the Connections panel (lightning bolt icon at bottom).

From	To	Cable Type
CR-1 Fa0/0	Dist-SW-1 Gi0/1	Straight-through Copper
CR-1 Fa0/1	Dist-SW-2 Gi0/1	Straight-through Copper
CR-1 Se0/0/0	ER-1 Se0/0/0	Serial DCE/DTE (Serial)
Dist-SW-1 Fa0/1	Access-SW-1 Fa0/1 (uplink)	Straight-through
Dist-SW-1 Fa0/2	Access-SW-2 Fa0/1 (uplink)	Straight-through

Dist-SW-2 Fa0/1	Access-SW-3 Fa0/1 (uplink)	Straight-through
Dist-SW-2 Fa0/2	Access-SW-4 Fa0/1 (uplink)	Straight-through
Access-SW-1 Fa0/2	Admin-PC1	Straight-through
Access-SW-1 Fa0/3	Admin-PC2	Straight-through
Access-SW-2 Fa0/2	IT-PC1	Straight-through
Access-SW-2 Fa0/3	IT-PC2	Straight-through
Access-SW-3 Fa0/2	Lab-PC1	Straight-through
Access-SW-3 Fa0/3	Lab-PC2	Straight-through
Access-SW-4 Fa0/2	Lib-PC1	Straight-through
Dist-SW-2 Fa0/5	TFTP-Server	Straight-through
Dist-SW-2 Fa0/6	DNS/Web Server	Straight-through
Dist-SW-1 Fa0/5	Mgmt-PC	Straight-through

IMPORTANT:
Serial links between CR-1 and ER-1 require a Serial WAN card module.
To add: Click on the router > Physical tab > drag WIC-2T module into empty slot > power on.
After adding the WIC-2T module, use interface Serial0/0/0 for both routers.
Set clock rate on the DCE side (CR-1 Se0/0/0): 'clock rate 64000'

4. End Device IP Configuration

Configure static IP addresses on all PCs and servers. Click on each device > Desktop tab > IP Configuration.

4.1 TFTP Server Configuration

The TFTP Server is the central component for all backup and IOS upgrade operations.

Parameter	Value
IP Address	192.168.100.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.100.1
DNS Server	192.168.100.20

To enable TFTP service: Click Server > Services tab > TFTP > toggle ON.

4.2 DNS / Web Server Configuration

Parameter	Value
IP Address	192.168.100.20
Subnet Mask	255.255.255.0
Default Gateway	192.168.100.1
Services	DNS: ON, HTTP: ON

DNS entry to add: 'university.local' -> 192.168.100.20 (Type A record)

4.3 Management PC

Parameter	Value
IP Address	192.168.50.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.50.1
DNS Server	192.168.100.20

4.4 Department PCs

Device	IP Address	Subnet Mask	Default Gateway
Admin-PC1	192.168.10.10	255.255.255.0	192.168.10.1
Admin-PC2	192.168.10.11	255.255.255.0	192.168.10.1
IT-PC1	192.168.20.10	255.255.255.0	192.168.20.1
IT-PC2	192.168.20.11	255.255.255.0	192.168.20.1
Lab-PC1	192.168.30.10	255.255.255.0	192.168.30.1
Lab-PC2	192.168.30.11	255.255.255.0	192.168.30.1
Lib-PC1	192.168.40.10	255.255.255.0	192.168.40.1

5. Switch Configuration

5.1 Distribution Switch 1 (Dist-SW-1) — L3 Switch

Dist-SW-1 handles VLANs 10 (Admin), 20 (IT), 50 (Management). It routes between these VLANs and has an L3 uplink to the Core Router.

STEP 1 Enter Global Configuration Mode

Dist-SW-1 — Initial Setup & Hostname

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Dist-SW-1
Dist-SW-1(config)# no ip domain-lookup
Dist-SW-1(config)# enable secret cisco123
Dist-SW-1(config)# service password-encryption
Dist-SW-1(config)# banner motd #
=== AUTHORIZED ACCESS ONLY - Dist-SW-1 ===
#
```

STEP 2 Enable IP Routing on L3 Switch

Dist-SW-1 — Enable Routing

```
Dist-SW-1(config)# ip routing
! Verify routing is enabled
Dist-SW-1(config)# do show ip route
```

STEP 3 Create VLANs

Dist-SW-1 — VLAN Creation

```
Dist-SW-1(config)# vlan 10
Dist-SW-1(config-vlan)# name Admin
Dist-SW-1(config-vlan)# exit
Dist-SW-1(config)# vlan 20
Dist-SW-1(config-vlan)# name IT
Dist-SW-1(config-vlan)# exit
Dist-SW-1(config)# vlan 50
Dist-SW-1(config-vlan)# name Management
Dist-SW-1(config-vlan)# exit
Dist-SW-1(config)# vlan 99
Dist-SW-1(config-vlan)# name Native
Dist-SW-1(config-vlan)# exit
```

STEP 4 Configure SVI (Switch Virtual Interfaces) for Inter-VLAN Routing

Dist-SW-1 — SVI Configuration

```
Dist-SW-1(config)# interface vlan 10
```

```
Dist-SW-1(config-if)# description Admin_VLAN_Gateway
Dist-SW-1(config-if)# ip address 192.168.10.1 255.255.255.0
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit

Dist-SW-1(config)# interface vlan 20
Dist-SW-1(config-if)# description IT_VLAN_Gateway
Dist-SW-1(config-if)# ip address 192.168.20.1 255.255.255.0
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit

Dist-SW-1(config)# interface vlan 50
Dist-SW-1(config-if)# description Management_VLAN_Gateway
Dist-SW-1(config-if)# ip address 192.168.50.1 255.255.255.0
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit
```

STEP 5 Configure Routed Uplink Port to Core Router

Dist-SW-1 — Uplink to CR-1

```
Dist-SW-1(config)# interface GigabitEthernet0/1
Dist-SW-1(config-if)# description Uplink_to_CR-1
Dist-SW-1(config-if)# no switchport
Dist-SW-1(config-if)# ip address 10.0.1.2 255.255.255.252
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit
```

STEP 6 Configure Trunk Links to Access Switches

Dist-SW-1 — Trunk Ports (Fa0/1 to Access-SW-1, Fa0/2 to Access-SW-2)

```
Dist-SW-1(config)# interface FastEthernet0/1
Dist-SW-1(config-if)# description Trunk_to_Access-SW-1
Dist-SW-1(config-if)# switchport mode trunk
Dist-SW-1(config-if)# switchport trunk native vlan 99
Dist-SW-1(config-if)# switchport trunk allowed vlan 10,20,50,99
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit

Dist-SW-1(config)# interface FastEthernet0/2
Dist-SW-1(config-if)# description Trunk_to_Access-SW-2
Dist-SW-1(config-if)# switchport mode trunk
Dist-SW-1(config-if)# switchport trunk native vlan 99
Dist-SW-1(config-if)# switchport trunk allowed vlan 10,20,50,99
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit
```

STEP 7 Configure Management PC Access Port

Dist-SW-1 — Mgmt-PC Access Port (Fa0/5)

```
Dist-SW-1(config)# interface FastEthernet0/5
Dist-SW-1(config-if)# description Mgmt-PC
```

```
Dist-SW-1(config-if)# switchport mode access
Dist-SW-1(config-if)# switchport access vlan 50
Dist-SW-1(config-if)# spanning-tree portfast
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit
```

STEP 8 Configure OSPF on Dist-SW-1

Dist-SW-1 — OSPF Configuration

```
Dist-SW-1(config)# router ospf 1
Dist-SW-1(config-router)# router-id 1.1.1.2
Dist-SW-1(config-router)# network 192.168.10.0 0.0.0.255 area 1
Dist-SW-1(config-router)# network 192.168.20.0 0.0.0.255 area 1
Dist-SW-1(config-router)# network 192.168.50.0 0.0.0.255 area 1
Dist-SW-1(config-router)# network 10.0.1.0 0.0.0.3 area 0
Dist-SW-1(config-router)# passive-interface vlan 10
Dist-SW-1(config-router)# passive-interface vlan 20
Dist-SW-1(config-router)# passive-interface vlan 50
Dist-SW-1(config-router)# exit
```

STEP 9 Configure SSH for Remote Management

Dist-SW-1 — SSH Configuration

```
Dist-SW-1(config)# ip domain-name university.local
Dist-SW-1(config)# crypto key generate rsa
  How many bits in the modulus [512]: 1024
Dist-SW-1(config)# ip ssh version 2
Dist-SW-1(config)# username admin privilege 15 secret Admin@123
Dist-SW-1(config)# line vty 0 4
Dist-SW-1(config-line)# transport input ssh
Dist-SW-1(config-line)# login local
Dist-SW-1(config-line)# exec-timeout 10 0
Dist-SW-1(config-line)# exit
Dist-SW-1(config)# line console 0
Dist-SW-1(config-line)# password console123
Dist-SW-1(config-line)# login
Dist-SW-1(config-line)# exec-timeout 10 0
Dist-SW-1(config-line)# exit
```

STEP 10 Save Configuration

Dist-SW-1 — Save

```
Dist-SW-1(config)# end
Dist-SW-1# write memory
! OR: Dist-SW-1# copy running-config startup-config
Destination filename [startup-config]?
[Press Enter]
Building configuration...
[OK]
```

5.2 Distribution Switch 2 (Dist-SW-2) — L3 Switch

Dist-SW-2 handles VLANs 30 (Labs), 40 (Library), 100 (Servers — TFTP, DNS). It connects to Access-SW-3, Access-SW-4, TFTP-Server, and DNS/Web Server.

Dist-SW-2 — Complete Configuration

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Dist-SW-2
Dist-SW-2(config)# no ip domain-lookup
Dist-SW-2(config)# enable secret cisco123
Dist-SW-2(config)# service password-encryption
Dist-SW-2(config)# ip routing

! --- VLAN Creation ---
Dist-SW-2(config)# vlan 30
Dist-SW-2(config-vlan)# name Labs
Dist-SW-2(config-vlan)# exit
Dist-SW-2(config)# vlan 40
Dist-SW-2(config-vlan)# name Library
Dist-SW-2(config-vlan)# exit
Dist-SW-2(config)# vlan 99
Dist-SW-2(config-vlan)# name Native
Dist-SW-2(config-vlan)# exit
Dist-SW-2(config)# vlan 100
Dist-SW-2(config-vlan)# name Servers
Dist-SW-2(config-vlan)# exit

! --- SVI for Inter-VLAN Routing ---
Dist-SW-2(config)# interface vlan 30
Dist-SW-2(config-if)# description Labs_VLAN_Gateway
Dist-SW-2(config-if)# ip address 192.168.30.1 255.255.255.0
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit

Dist-SW-2(config)# interface vlan 40
Dist-SW-2(config-if)# description Library_VLAN_Gateway
Dist-SW-2(config-if)# ip address 192.168.40.1 255.255.255.0
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit

Dist-SW-2(config)# interface vlan 100
Dist-SW-2(config-if)# description Servers_VLAN_Gateway
Dist-SW-2(config-if)# ip address 192.168.100.1 255.255.255.0
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit

! --- Uplink to CR-1 (L3 Routed Port) ---
Dist-SW-2(config)# interface GigabitEthernet0/1
Dist-SW-2(config-if)# description Uplink_to_CR-1
Dist-SW-2(config-if)# no switchport
Dist-SW-2(config-if)# ip address 10.0.2.2 255.255.255.252
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit

! --- Trunk Ports to Access Switches ---
Dist-SW-2(config)# interface FastEthernet0/1
Dist-SW-2(config-if)# description Trunk_to_Access-SW-3
Dist-SW-2(config-if)# switchport mode trunk
```

```
Dist-SW-2(config-if)# switchport trunk native vlan 99
Dist-SW-2(config-if)# switchport trunk allowed vlan 30,40,99,100
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit

Dist-SW-2(config)# interface FastEthernet0/2
Dist-SW-2(config-if)# description Trunk_to_Access-SW-4
Dist-SW-2(config-if)# switchport mode trunk
Dist-SW-2(config-if)# switchport trunk native vlan 99
Dist-SW-2(config-if)# switchport trunk allowed vlan 30,40,99,100
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit

! --- Server Access Ports ---
Dist-SW-2(config)# interface FastEthernet0/5
Dist-SW-2(config-if)# description TFTP-Server
Dist-SW-2(config-if)# switchport mode access
Dist-SW-2(config-if)# switchport access vlan 100
Dist-SW-2(config-if)# spanning-tree portfast
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit

Dist-SW-2(config)# interface FastEthernet0/6
Dist-SW-2(config-if)# description DNS-Web-Server
Dist-SW-2(config-if)# switchport mode access
Dist-SW-2(config-if)# switchport access vlan 100
Dist-SW-2(config-if)# spanning-tree portfast
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit

! --- OSPF ---
Dist-SW-2(config)# router ospf 1
Dist-SW-2(config-router)# router-id 1.1.1.3
Dist-SW-2(config-router)# network 192.168.30.0 0.0.0.255 area 2
Dist-SW-2(config-router)# network 192.168.40.0 0.0.0.255 area 2
Dist-SW-2(config-router)# network 192.168.100.0 0.0.0.255 area 2
Dist-SW-2(config-router)# network 10.0.2.0 0.0.0.3 area 0
Dist-SW-2(config-router)# passive-interface vlan 30
Dist-SW-2(config-router)# passive-interface vlan 40
Dist-SW-2(config-router)# passive-interface vlan 100
Dist-SW-2(config-router)# exit

! --- SSH ---
Dist-SW-2(config)# ip domain-name university.local
Dist-SW-2(config)# crypto key generate rsa
  How many bits in the modulus [512]: 1024
Dist-SW-2(config)# ip ssh version 2
Dist-SW-2(config)# username admin privilege 15 secret Admin@123
Dist-SW-2(config)# line vty 0 4
Dist-SW-2(config-line)# transport input ssh
Dist-SW-2(config-line)# login local
Dist-SW-2(config-line)# exit

Dist-SW-2(config)# end
Dist-SW-2# write memory
```

5.3 Access Switch 1 (Access-SW-1) — Admin Department

Access-SW-1 — Complete Configuration

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Access-SW-1
Access-SW-1(config)# no ip domain-lookup
Access-SW-1(config)# enable secret cisco123
Access-SW-1(config)# service password-encryption

! --- VLAN Creation ---
Access-SW-1(config)# vlan 10
Access-SW-1(config-vlan)# name Admin
Access-SW-1(config-vlan)# exit
Access-SW-1(config)# vlan 99
Access-SW-1(config-vlan)# name Native
Access-SW-1(config-vlan)# exit

! --- Uplink Trunk to Dist-SW-1 ---
Access-SW-1(config)# interface FastEthernet0/1
Access-SW-1(config-if)# description Trunk_to_Dist-SW-1
Access-SW-1(config-if)# switchport mode trunk
Access-SW-1(config-if)# switchport trunk native vlan 99
Access-SW-1(config-if)# switchport trunk allowed vlan 10,99
Access-SW-1(config-if)# no shutdown
Access-SW-1(config-if)# exit

! --- Access Ports for Admin PCs ---
Access-SW-1(config)# interface FastEthernet0/2
Access-SW-1(config-if)# description Admin-PC1
Access-SW-1(config-if)# switchport mode access
Access-SW-1(config-if)# switchport access vlan 10
Access-SW-1(config-if)# switchport port-security
Access-SW-1(config-if)# switchport port-security maximum 1
Access-SW-1(config-if)# switchport port-security violation restrict
Access-SW-1(config-if)# switchport port-security mac-address sticky
Access-SW-1(config-if)# spanning-tree portfast
Access-SW-1(config-if)# no shutdown
Access-SW-1(config-if)# exit

Access-SW-1(config)# interface FastEthernet0/3
Access-SW-1(config-if)# description Admin-PC2
Access-SW-1(config-if)# switchport mode access
Access-SW-1(config-if)# switchport access vlan 10
Access-SW-1(config-if)# switchport port-security
Access-SW-1(config-if)# switchport port-security maximum 1
Access-SW-1(config-if)# switchport port-security violation restrict
Access-SW-1(config-if)# switchport port-security mac-address sticky
Access-SW-1(config-if)# spanning-tree portfast
Access-SW-1(config-if)# no shutdown
Access-SW-1(config-if)# exit

! --- Management SVI ---
Access-SW-1(config)# interface vlan 10
Access-SW-1(config-if)# ip address 192.168.10.250 255.255.255.0
Access-SW-1(config-if)# no shutdown
Access-SW-1(config-if)# exit
Access-SW-1(config)# ip default-gateway 192.168.10.1
```

```
! --- Disable unused ports ---
Access-SW-1(config)# interface range FastEthernet0/4-24
Access-SW-1(config-if-range)# shutdown
Access-SW-1(config-if-range)# exit

Access-SW-1(config)# end
Access-SW-1# write memory
```

5.4 Access Switch 2 (Access-SW-2) — IT Department

Access-SW-2 — Complete Configuration

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Access-SW-2
Access-SW-2(config)# no ip domain-lookup
Access-SW-2(config)# enable secret cisco123
Access-SW-2(config)# service password-encryption

Access-SW-2(config)# vlan 20
Access-SW-2(config-vlan)# name IT
Access-SW-2(config-vlan)# exit
Access-SW-2(config)# vlan 99
Access-SW-2(config-vlan)# name Native
Access-SW-2(config-vlan)# exit

Access-SW-2(config)# interface FastEthernet0/1
Access-SW-2(config-if)# description Trunk_to_Dist-SW-1
Access-SW-2(config-if)# switchport mode trunk
Access-SW-2(config-if)# switchport trunk native vlan 99
Access-SW-2(config-if)# switchport trunk allowed vlan 20,99
Access-SW-2(config-if)# no shutdown
Access-SW-2(config-if)# exit

Access-SW-2(config)# interface FastEthernet0/2
Access-SW-2(config-if)# description IT-PC1
Access-SW-2(config-if)# switchport mode access
Access-SW-2(config-if)# switchport access vlan 20
Access-SW-2(config-if)# switchport port-security
Access-SW-2(config-if)# switchport port-security maximum 1
Access-SW-2(config-if)# switchport port-security violation restrict
Access-SW-2(config-if)# switchport port-security mac-address sticky
Access-SW-2(config-if)# spanning-tree portfast
Access-SW-2(config-if)# no shutdown
Access-SW-2(config-if)# exit

Access-SW-2(config)# interface FastEthernet0/3
Access-SW-2(config-if)# description IT-PC2
Access-SW-2(config-if)# switchport mode access
Access-SW-2(config-if)# switchport access vlan 20
Access-SW-2(config-if)# switchport port-security
Access-SW-2(config-if)# switchport port-security maximum 1
Access-SW-2(config-if)# switchport port-security violation restrict
Access-SW-2(config-if)# switchport port-security mac-address sticky
Access-SW-2(config-if)# spanning-tree portfast
Access-SW-2(config-if)# no shutdown
Access-SW-2(config-if)# exit
```

```
Access-SW-2(config)# interface vlan 20
Access-SW-2(config-if)# ip address 192.168.20.250 255.255.255.0
Access-SW-2(config-if)# no shutdown
Access-SW-2(config-if)# exit
Access-SW-2(config)# ip default-gateway 192.168.20.1

Access-SW-2(config)# interface range FastEthernet0/4-24
Access-SW-2(config-if-range)# shutdown
Access-SW-2(config-if-range)# exit
Access-SW-2(config)# end
Access-SW-2# write memory
```

5.5 Access Switch 3 (Access-SW-3) — Computer Labs

Access-SW-3 — Complete Configuration

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Access-SW-3
Access-SW-3(config)# no ip domain-lookup
Access-SW-3(config)# enable secret cisco123
Access-SW-3(config)# service password-encryption

Access-SW-3(config)# vlan 30
Access-SW-3(config-vlan)# name Labs
Access-SW-3(config-vlan)# exit
Access-SW-3(config)# vlan 99
Access-SW-3(config-vlan)# name Native
Access-SW-3(config-vlan)# exit

Access-SW-3(config)# interface FastEthernet0/1
Access-SW-3(config-if)# description Trunk_to_Dist-SW-2
Access-SW-3(config-if)# switchport mode trunk
Access-SW-3(config-if)# switchport trunk native vlan 99
Access-SW-3(config-if)# switchport trunk allowed vlan 30,99
Access-SW-3(config-if)# no shutdown
Access-SW-3(config-if)# exit

Access-SW-3(config)# interface FastEthernet0/2
Access-SW-3(config-if)# description Lab-PC1
Access-SW-3(config-if)# switchport mode access
Access-SW-3(config-if)# switchport access vlan 30
Access-SW-3(config-if)# spanning-tree portfast
Access-SW-3(config-if)# no shutdown
Access-SW-3(config-if)# exit

Access-SW-3(config)# interface FastEthernet0/3
Access-SW-3(config-if)# description Lab-PC2
Access-SW-3(config-if)# switchport mode access
Access-SW-3(config-if)# switchport access vlan 30
Access-SW-3(config-if)# spanning-tree portfast
Access-SW-3(config-if)# no shutdown
Access-SW-3(config-if)# exit

Access-SW-3(config)# interface vlan 30
```

```
Access-SW-3(config-if)# ip address 192.168.30.250 255.255.255.0
Access-SW-3(config-if)# no shutdown
Access-SW-3(config-if)# exit
Access-SW-3(config)# ip default-gateway 192.168.30.1

Access-SW-3(config)# interface range FastEthernet0/4-24
Access-SW-3(config-if-range)# shutdown
Access-SW-3(config-if-range)# exit
Access-SW-3(config)# end
Access-SW-3# write memory
```

5.6 Access Switch 4 (Access-SW-4) — Library

Access-SW-4 — Complete Configuration

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Access-SW-4
Access-SW-4(config)# no ip domain-lookup
Access-SW-4(config)# enable secret cisco123
Access-SW-4(config)# service password-encryption

Access-SW-4(config)# vlan 40
Access-SW-4(config-vlan)# name Library
Access-SW-4(config-vlan)# exit
Access-SW-4(config)# vlan 99
Access-SW-4(config-vlan)# name Native
Access-SW-4(config-vlan)# exit

Access-SW-4(config)# interface FastEthernet0/1
Access-SW-4(config-if)# description Trunk_to_Dist-SW-2
Access-SW-4(config-if)# switchport mode trunk
Access-SW-4(config-if)# switchport trunk native vlan 99
Access-SW-4(config-if)# switchport trunk allowed vlan 40,99
Access-SW-4(config-if)# no shutdown
Access-SW-4(config-if)# exit

Access-SW-4(config)# interface FastEthernet0/2
Access-SW-4(config-if)# description Lib-PC1
Access-SW-4(config-if)# switchport mode access
Access-SW-4(config-if)# switchport access vlan 40
Access-SW-4(config-if)# spanning-tree portfast
Access-SW-4(config-if)# no shutdown
Access-SW-4(config-if)# exit

Access-SW-4(config)# interface vlan 40
Access-SW-4(config-if)# ip address 192.168.40.250 255.255.255.0
Access-SW-4(config-if)# no shutdown
Access-SW-4(config-if)# exit
Access-SW-4(config)# ip default-gateway 192.168.40.1

Access-SW-4(config)# interface range FastEthernet0/3-24
Access-SW-4(config-if-range)# shutdown
Access-SW-4(config-if-range)# exit
Access-SW-4(config)# end
Access-SW-4# write memory
```

6. Router Configuration

6.1 Core Router (CR-1)

CR-1 is the backbone router. It runs OSPF Area 0 (backbone area), connecting to both distribution switches and the edge router via a serial WAN link.

CR-1 — Complete Configuration

```
Router> enable
Router# configure terminal
Router(config)# hostname CR-1
CR-1(config)# no ip domain-lookup
CR-1(config)# enable secret cisco123
CR-1(config)# service password-encryption
CR-1(config)# banner motd #
=== AUTHORIZED ACCESS ONLY - Core Router CR-1 ===
#

! --- Interface to Dist-SW-1 ---
CR-1(config)# interface FastEthernet0/0
CR-1(config-if)# description Link_to_Dist-SW-1
CR-1(config-if)# ip address 10.0.1.1 255.255.255.252
CR-1(config-if)# no shutdown
CR-1(config-if)# exit

! --- Interface to Dist-SW-2 ---
CR-1(config)# interface FastEthernet0/1
CR-1(config-if)# description Link_to_Dist-SW-2
CR-1(config-if)# ip address 10.0.2.1 255.255.255.252
CR-1(config-if)# no shutdown
CR-1(config-if)# exit

! --- Serial WAN Link to Edge Router ---
CR-1(config)# interface Serial0/0/0
CR-1(config-if)# description WAN_Link_to_ER-1
CR-1(config-if)# ip address 10.0.0.1 255.255.255.252
CR-1(config-if)# clock rate 64000
CR-1(config-if)# no shutdown
CR-1(config-if)# exit

! --- OSPF Configuration ---
CR-1(config)# router ospf 1
CR-1(config-router)# router-id 1.1.1.1
CR-1(config-router)# network 10.0.0.0 0.0.0.3 area 0
CR-1(config-router)# network 10.0.1.0 0.0.0.3 area 0
CR-1(config-router)# network 10.0.2.0 0.0.0.3 area 0
CR-1(config-router)# exit

! --- SSH Configuration ---
CR-1(config)# ip domain-name university.local
CR-1(config)# crypto key generate rsa
  How many bits in the modulus [512]: 1024
CR-1(config)# ip ssh version 2
CR-1(config)# username admin privilege 15 secret Admin@123
CR-1(config)# line vty 0 4
CR-1(config-line)# transport input ssh
CR-1(config-line)# login local
CR-1(config-line)# exec-timeout 10 0
```

```
CR-1(config-line)# exit
CR-1(config)# line console 0
CR-1(config-line)# password console123
CR-1(config-line)# login
CR-1(config-line)# exit

CR-1(config)# end
CR-1# write memory
```

6.2 Edge Router (ER-1)

ER-1 acts as the WAN edge device. It connects to CR-1 via serial link and simulates an ISP/Internet connection on its Fa0/0. It also runs OSPF and distributes a default route.

ER-1 — Complete Configuration

```
Router> enable
Router# configure terminal
Router(config)# hostname ER-1
ER-1(config)# no ip domain-lookup
ER-1(config)# enable secret cisco123
ER-1(config)# service password-encryption
ER-1(config)# banner motd #
=== AUTHORIZED ACCESS ONLY - Edge Router ER-1 ===
#

! --- WAN Interface (to ISP/Internet - simulated) ---
ER-1(config)# interface FastEthernet0/0
ER-1(config-if)# description Internet_Facing
ER-1(config-if)# ip address 203.0.113.1 255.255.255.0
ER-1(config-if)# ip nat outside
ER-1(config-if)# no shutdown
ER-1(config-if)# exit

! --- Serial Link to CR-1 ---
ER-1(config)# interface Serial0/0/0
ER-1(config-if)# description WAN_Link_to_CR-1
ER-1(config-if)# ip address 10.0.0.2 255.255.255.252
ER-1(config-if)# ip nat inside
ER-1(config-if)# no shutdown
ER-1(config-if)# exit

! --- Default Route to Internet ---
ER-1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.254

! --- OSPF - Redistribute Default Route ---
ER-1(config)# router ospf 1
ER-1(config-router)# router-id 1.1.1.4
ER-1(config-router)# network 10.0.0.0 0.0.0.3 area 0
ER-1(config-router)# default-information originate
ER-1(config-router)# exit

! --- NAT Configuration ---
! Define which internal traffic gets NATed
ER-1(config)# access-list 10 permit 192.168.0.0 0.0.255.255
ER-1(config)# ip nat inside source list 10 interface FastEthernet0/0 overload
```

```
! --- SSH ---
ER-1(config)# ip domain-name university.local
ER-1(config)# crypto key generate rsa
  How many bits in the modulus [512]: 1024
ER-1(config)# ip ssh version 2
ER-1(config)# username admin privilege 15 secret Admin@123
ER-1(config)# line vty 0 4
ER-1(config-line)# transport input ssh
ER-1(config-line)# login local
ER-1(config-line)# exit

ER-1(config)# end
ER-1# write memory
```

7. OSPF Verification & Connectivity Testing

7.1 Verify OSPF Neighbor Relationships

Run the following commands on each router/L3 switch after all configurations are complete. Allow 1-2 minutes for OSPF to fully converge in CPT.

OSPF Verification Commands (run on CR-1, Dist-SW-1, Dist-SW-2)

```
! Check OSPF neighbors
CR-1# show ip ospf neighbor

! Expected output: Should show 3 neighbors (ER-1, Dist-SW-1, Dist-SW-2)
Neighbor ID    Pri   State       Dead Time   Address     Interface
1.1.1.4        1     FULL/DR     00:00:38    10.0.0.2    Serial0/0/0
1.1.1.2        1     FULL/DR     00:00:34    10.0.1.2    FastEthernet0/0
1.1.1.3        1     FULL/DR     00:00:36    10.0.2.2    FastEthernet0/1

! Check OSPF routing table
CR-1# show ip route ospf

! Check OSPF database
CR-1# show ip ospf database

! Check specific route
CR-1# show ip route 192.168.10.0
```

7.2 End-to-End Connectivity Tests

Use the Simulation mode in Packet Tracer or the CLI ping command to verify connectivity.

Ping Tests from Admin-PC1 (192.168.10.10)

```
! Test 1: Ping default gateway
C:\> ping 192.168.10.1

! Test 2: Ping another VLAN (IT)
C:\> ping 192.168.20.10

! Test 3: Ping TFTP Server
C:\> ping 192.168.100.10

! Test 4: Ping across serial WAN link
C:\> ping 10.0.0.2

! Test 5: Traceroute to verify path
C:\> tracert 192.168.30.10
```

8. TFTP Configuration Backup

This is the core feature of the project. The TFTP server (192.168.100.10) will store backed-up running and startup configurations from all devices.

8.1 How TFTP Works in CPT

- TFTP (Trivial File Transfer Protocol) uses UDP port 69
- No authentication — relies on network-level security (ACLs)
- In Packet Tracer, the TFTP server must have the TFTP service enabled
- Files are stored on the server and can be retrieved later for restore

8.2 Enable TFTP Service on Server

1. Click on the TFTP-Server in CPT
2. Go to the Services tab
3. Click TFTP on the left panel
4. Toggle the service to ON
5. Note the directory listing — uploaded files will appear here

8.3 Backup Running Configuration from CR-1

CR-1 — Backup Running-Config to TFTP

```
CR-1# copy running-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-confg]? CR-1-running-backup

Writing running-config...!!
[OK - 1234 bytes]

! Verify the file was transferred
CR-1# show flash:
```

8.4 Backup Startup Configuration from CR-1

CR-1 — Backup Startup-Config to TFTP

```
CR-1# copy startup-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-confg]? CR-1-startup-backup

Writing startup-config...!!
[OK - 1234 bytes]
```

8.5 Backup All Devices — Command Reference

Repeat the backup process for every device. Use a consistent naming convention.

Device	Command	TFTP Server IP	Filename
CR-1	copy running-config tftp	192.168.100.10	CR-1-running-backup
ER-1	copy running-config tftp	192.168.100.10	ER-1-running-backup
Dist-SW-1	copy running-config tftp	192.168.100.10	DistSW1-running-backup
Dist-SW-2	copy running-config tftp	192.168.100.10	DistSW2-running-backup
Access-SW-1	copy running-config tftp	192.168.100.10	AccSW1-running-backup
Access-SW-2	copy running-config tftp	192.168.100.10	AccSW2-running-backup
Access-SW-3	copy running-config tftp	192.168.100.10	AccSW3-running-backup
Access-SW-4	copy running-config tftp	192.168.100.10	AccSW4-running-backup

8.6 Verify Backup on TFTP Server

After backups, open the TFTP Server > Services > TFTP. You should see all backup files listed in the file table. Each file will show its name and size.

> **NOTE:**
> In Packet Tracer 9.0, TFTP file names are case-sensitive.
> The TFTP server GUI under Services > TFTP shows all stored files.
> You cannot open/edit files directly in CPT — they are stored as binary blobs.

8.7 Restore Configuration from TFTP

To restore a backed-up configuration from TFTP to a device:

CR-1 — Restore Running-Config from TFTP

```
! Method 1: Restore directly to running-config (immediate effect)
CR-1# copy tftp running-config
Address or name of remote host []? 192.168.100.10
Source filename []? CR-1-running-backup
Destination filename [running-config]?
[Press Enter]

Accessing tftp://192.168.100.10/CR-1-running-backup...
Loading CR-1-running-backup from 192.168.100.10 (via FastEthernet0/0):
!
[OK - 1234 bytes]

! Method 2: Restore to startup-config (takes effect after reload)
CR-1# copy tftp startup-config
```

```
Address or name of remote host []? 192.168.100.10
Source filename []? CR-1-startup-backup
Destination filename [startup-config]?
[Press Enter]

[OK - 1234 bytes]
CR-1# reload
Proceed with reload? [confirm]
[Press Enter]
```

9. IOS Image Backup and Upgrade via TFTP

9.1 Understanding IOS Image Management

In a production environment, IOS images are large binary files (several MB to hundreds of MB). In Cisco Packet Tracer, IOS images are simulated — the process is demonstrated but actual file sizes and real upgrade behavior may differ from physical hardware.

- IOS image filename format: [platform]-[feature_set]-[format].[version].bin
- Example: c2900-universalk9-mz.SPA.151-4.M4.bin
- TFTP is the traditional method for IOS transfers in enterprise environments
- In CPT, the 'show version' command displays the simulated IOS version

9.2 Step 1 — Check Current IOS Version

CR-1 — Verify Current IOS Version

```
CR-1# show version
Cisco IOS Software, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
...
ROM: System Bootstrap, Version 15.0(1)M3
...
System image file is 'flash:/c2900-universalk9-mz.SPA.151-4.M4.bin'

! Check flash memory contents
CR-1# show flash:
System flash directory:
File  Length    Name/status
  1   33591768  c2900-universalk9-mz.SPA.151-4.M4.bin
[33591768 bytes used, 221716384 available, 255308152 total]
255M bytes of processor board System flash (Read/Write)
```

9.3 Step 2 — Back Up Current IOS to TFTP

Always back up the existing IOS image before attempting an upgrade. This ensures you can recover if the upgrade fails.

CR-1 — Backup IOS Image to TFTP Server

```
CR-1# copy flash: tftp:
Source filename []? c2900-universalk9-mz.SPA.151-4.M4.bin
Address or name of remote host []? 192.168.100.10
Destination filename [c2900-universalk9-mz.SPA.151-4.M4.bin]?
  CR-1-IOS-backup.bin

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[OK - 33591768 bytes]

! Verify on server: go to TFTP Server > Services > TFTP
```

```
! You should see 'CR-1-IOS-backup.bin' in the file list
```

9.4 Step 3 — Simulate Loading New IOS from TFTP

In Packet Tracer, we simulate the IOS upgrade by copying an IOS image from the TFTP server to flash. In CPT, this demonstrates the process even if the image doesn't actually change the running version.

CR-1 — Download New IOS from TFTP to Flash

```
! First, verify available flash space
CR-1# show flash:

! Copy new IOS from TFTP to flash
CR-1# copy tftp flash:
Address or name of remote host []? 192.168.100.10
Source filename []? c2900-universalk9-mz.SPA.155-3.M.bin
Destination filename [c2900-universalk9-mz.SPA.155-3.M.bin]?
[Press Enter]

Accessing tftp://192.168.100.10/c2900-universalk9-mz.SPA.155-3.M.bin...
Loading c2900-universalk9-mz.SPA.155-3.M.bin ...
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[OK - 33591768 bytes]

Verifying checksum... OK (0x9E4A)
```

9.5 Step 4 — Configure Boot System for New IOS

After copying the new image, configure the router to boot from it on the next reload.

CR-1 — Set Boot System & Reload

```
CR-1# configure terminal
! Set boot order: try new image first, fall back to old
CR-1(config)# boot system flash:c2900-universalk9-mz.SPA.155-3.M.bin
CR-1(config)# boot system flash:c2900-universalk9-mz.SPA.151-4.M4.bin
CR-1(config)# end

! Save configuration before reload
CR-1# write memory

! Verify boot configuration
CR-1# show boot
BOOT path-list: flash:c2900-universalk9-mz.SPA.155-3.M.bin
                flash:c2900-universalk9-mz.SPA.151-4.M4.bin

! Reload the router
CR-1# reload
System configuration has been modified. Save? [yes/no]: no
Proceed with reload? [confirm]
```

```
[Press Enter]

! After reload, verify new IOS version
CR-1# show version
```

CPT NOTE:
In Cisco Packet Tracer, the IOS image file used for simulation is built into the software — not a separate file. The 'copy tftp flash' process is simulated. The 'boot system' and reload process IS functional in CPT and will demonstrate the upgrade workflow. For IOS Upgrade simulation, use files already in CPT flash.

10. Advanced Configuration & Security

10.1 Access Control Lists (ACLs)

ACLs restrict which devices can access the TFTP server and management interfaces, improving security.

CR-1 — ACL: Restrict TFTP Access to Management VLAN Only

```
CR-1(config)# ip access-list extended TFTP-ACCESS
CR-1(config-ext-nacl)# remark -- Allow Management VLAN to TFTP server --
CR-1(config-ext-nacl)# permit udp 192.168.50.0 0.0.0.255 host 192.168.100.10 eq
69
CR-1(config-ext-nacl)# remark -- Allow all internal VLANs to reach TFTP --
CR-1(config-ext-nacl)# permit udp 192.168.0.0 0.0.255.255 host 192.168.100.10
eq 69
CR-1(config-ext-nacl)# deny ip any host 192.168.100.10 log
CR-1(config-ext-nacl)# permit ip any any
CR-1(config-ext-nacl)# exit
```

10.2 Spanning Tree Protocol (STP) Optimization

Dist-SW-1 — STP Root Bridge Configuration

```
! Make Dist-SW-1 the root bridge for VLANs 10, 20, 50
Dist-SW-1(config)# spanning-tree vlan 10 root primary
Dist-SW-1(config)# spanning-tree vlan 20 root primary
Dist-SW-1(config)# spanning-tree vlan 50 root primary

! Make Dist-SW-2 the root bridge for VLANs 30, 40, 100
! (run on Dist-SW-2)
Dist-SW-2(config)# spanning-tree vlan 30 root primary
Dist-SW-2(config)# spanning-tree vlan 40 root primary
Dist-SW-2(config)# spanning-tree vlan 100 root primary

! Enable BPDU Guard on all access ports
Dist-SW-1(config)# spanning-tree portfast bpduguard default
```

10.3 DHCP Configuration on ER-1

Optionally, configure DHCP pools on ER-1 or Dist-SW-1 for dynamic IP assignment to end devices.

Dist-SW-1 — DHCP Pool for Admin VLAN

```
Dist-SW-1(config)# ip dhcp pool ADMIN-POOL
Dist-SW-1(dhcp-config)# network 192.168.10.0 255.255.255.0
Dist-SW-1(dhcp-config)# default-router 192.168.10.1
Dist-SW-1(dhcp-config)# dns-server 192.168.100.20
Dist-SW-1(dhcp-config)# lease 7
Dist-SW-1(dhcp-config)# exit
```

```
! Exclude static IPs from DHCP range
Dist-SW-1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.20

Dist-SW-1(config)# ip dhcp pool IT-POOL
Dist-SW-1(dhcp-config)# network 192.168.20.0 255.255.255.0
Dist-SW-1(dhcp-config)# default-router 192.168.20.1
Dist-SW-1(dhcp-config)# dns-server 192.168.100.20
Dist-SW-1(dhcp-config)# lease 7
Dist-SW-1(dhcp-config)# exit
Dist-SW-1(config)# ip dhcp excluded-address 192.168.20.1 192.168.20.20
```

11. Complete Verification Command Reference

11.1 Switch Verification

Switch — Verification Commands

```
! Verify VLANs
Dist-SW-1# show vlan brief

! Verify trunk links
Dist-SW-1# show interfaces trunk

! Verify SVI status
Dist-SW-1# show interfaces vlan 10
Dist-SW-1# show ip interface brief

! Verify port security
Access-SW-1# show port-security interface FastEthernet0/2
Access-SW-1# show port-security address

! Verify spanning tree
Dist-SW-1# show spanning-tree vlan 10

! Verify CDP neighbors
Dist-SW-1# show cdp neighbors
Dist-SW-1# show cdp neighbors detail
```

11.2 Router Verification

Router — Verification Commands

```
! Verify interface status
CR-1# show ip interface brief

! Verify OSPF
CR-1# show ip ospf neighbor
CR-1# show ip ospf
CR-1# show ip route ospf

! Verify routing table
CR-1# show ip route

! Verify SSH sessions
CR-1# show ssh
CR-1# show users

! Verify NAT translations (on ER-1)
ER-1# show ip nat translations
ER-1# show ip nat statistics

! Verify ACLs
CR-1# show access-lists
CR-1# show ip access-lists TFTP-ACCESS
```

11.3 TFTP Verification

TFTP Operations — Verification

```
! List files in flash
CR-1# show flash:

! Verify config was saved
CR-1# show startup-config

! Verify running config
CR-1# show running-config

! After TFTP copy, verify the transfer
CR-1# debug tftp
! (Run during transfer to see debug output)
CR-1# undebug all
! (Turn off debug after verification)
```

12. Using Simulation Mode in Packet Tracer

12.1 Setting Up Simulation

6. Click the Simulation tab in the bottom-right corner of CPT (looks like a stopwatch icon)
7. In the Event List Filters, select the protocols you want to trace: ICMP, TFTP, OSPF
8. From a PC, initiate a ping or TFTP operation
9. Click Play or Step to advance the simulation
10. Watch packets flow through the topology with colored envelopes

12.2 Simulating TFTP Backup

11. Switch to Simulation Mode
12. Add filter: TFTP (UDP/69)
13. Open CLI on CR-1 and type: copy running-config tftp
14. Enter TFTP server IP and filename
15. Click Play in Simulation panel
16. Observe TFTP packets traveling from CR-1 through the network to the TFTP Server
17. Click on any packet envelope to inspect PDU details

12.3 Simulating OSPF Convergence

18. Switch to Simulation Mode
19. Add filter: OSPF
20. Power cycle (shutdown/no shutdown) an interface on CR-1
21. Click Play to watch OSPF hello packets, LSAs, and reconvergence

12.4 Verifying Inter-VLAN Traffic Path

22. Switch to Simulation Mode
23. Add filter: ICMP
24. From Admin-PC1, ping Library-PC1 (192.168.40.10)
25. Observe: PC1 > Access-SW-1 > Dist-SW-1 > CR-1 > Dist-SW-2 > Access-SW-4 > Lib-PC1
26. Click on each packet to verify the IP headers and encapsulation at each hop

13. Troubleshooting Guide

13.1 Common Issues & Fixes

Problem	Solution / Check
OSPF neighbors not forming	Check IP addresses on P2P links (/30), verify 'no shutdown', check 'network' statements match interface subnets exactly
Inter-VLAN routing not working	Verify 'ip routing' is enabled on L3 switch, check SVIs are up/up, verify trunk allows correct VLANs
TFTP transfer fails	Check connectivity (ping TFTP server first), verify TFTP service is ON, check ACLs are not blocking UDP/69
VLANs not appearing on access switch	Ensure VLANs are created on access switch AND allowed on trunk, check 'show vlan brief'
SSH not working	Verify 'crypto key generate rsa', check 'ip ssh version 2', confirm VTY lines use 'transport input ssh'
Serial link not up	Verify WIC-2T module is installed, check 'clock rate' is set on DCE side, verify both interfaces are 'no shutdown'
Port security violation	Check 'show port-security', use 'shutdown' then 'no shutdown' on the port to clear violation
Ping fails across subnets	Verify default gateway on PCs is correct, check routing table has routes to destination, verify OSPF is converged

13.2 Debug Commands

Useful Debug Commands

```
! Debug OSPF adjacency issues
CR-1# debug ip ospf adj
CR-1# debug ip ospf events

! Debug TFTP transfers
CR-1# debug tftp

! Debug routing
CR-1# debug ip routing

! Debug ICMP (ping)
CR-1# debug ip icmp

! ALWAYS turn off debug when done!
CR-1# undebug all
! OR:
CR-1# no debug all
```

14. Project Summary & Expected Results

14.1 Expected Final State

Feature	Expected Result	Verification Command
OSPF Adjacencies	All routers and L3 switches show FULL/DR or FULL/BDR neighbors	show ip ospf neighbor
VLAN Routing	All VLANs can ping each other through L3 switches	ping across VLANs
TFTP Backup	All 8 device configs backed up on TFTP server	TFTP Server > TFTP files list
SSH Access	Admin can SSH from Mgmt-PC to all routers/switches	ssh -l admin <device IP>
Port Security	Access ports reject unauthorized MACs	show port-security
IOS Backup	Current IOS backed up on TFTP server	show flash:
STP	Dist-SW-1/SW-2 are root bridges for respective VLANs	show spanning-tree
NAT	Internal hosts can reach simulated Internet (203.0.113.x)	show ip nat translations

14.2 Project File Naming Convention

Save your Packet Tracer file as: UniversityNetwork_TFTP_Project.pkt

Item	Convention
Config Backup Files	DeviceName-running-backup, DeviceName-startup-backup
IOS Backup Files	DeviceName-IOS-backup.bin
OSPF Areas	Area 0 (Backbone): CR-1, ER-1 links | Area 1: Dist-SW-1 VLANs | Area 2: Dist-SW-2 VLANs
Management VLAN	VLAN 50: All device management IPs accessible from 192.168.50.10

14.3 Documentation Checklist

- Network topology diagram (screenshot from CPT with labels)
- IP addressing table — filled out for all devices
- VLAN table with port assignments
- Screenshot of OSPF neighbor table from CR-1
- Screenshot of TFTP server file list showing all backups
- Screenshot of successful ping across all VLANs

- Screenshot of SSH session from Mgmt-PC to CR-1
- Screenshot of 'show flash' showing IOS backup file
- CPT simulation mode screenshot showing TFTP packet flow

Appendix A: Quick Command Reference

Task	Command
Backup running-config to TFTP	copy running-config tftp
Backup startup-config to TFTP	copy startup-config tftp
Restore config from TFTP	copy tftp running-config
Backup IOS to TFTP	copy flash: tftp:
Copy IOS from TFTP	copy tftp flash:
Set boot IOS image	boot system flash:[filename]
Verify OSPF neighbors	show ip ospf neighbor
Verify routing table	show ip route
Verify VLANs	show vlan brief
Verify trunks	show interfaces trunk
Verify interface status	show ip interface brief
Verify flash contents	show flash:
Verify IOS version	show version
Verify port security	show port-security
Verify NAT	show ip nat translations
Save configuration	write memory
Enable routing (L3 switch)	ip routing
Generate RSA keys for SSH	crypto key generate rsa

Appendix B: Device Management IPs

Device	Mgmt IP	VLAN / Interface	Notes
CR-1	10.0.1.1	VLAN N/A	SSH: username admin
ER-1	10.0.0.2	VLAN N/A	SSH: username admin
Dist-SW-1	192.168.50.1	VLAN 50 SVI	SSH: username admin
Dist-SW-2	192.168.100.1	VLAN 100 SVI	SSH: username admin
Access-SW-1	192.168.10.250	VLAN 10 SVI	Default gateway: .1
Access-SW-2	192.168.20.250	VLAN 20 SVI	Default gateway: .1
Access-SW-3	192.168.30.250	VLAN 30 SVI	Default gateway: .1
Access-SW-4	192.168.40.250	VLAN 40 SVI	Default gateway: .1
TFTP-Server	192.168.100.10	VLAN 100	Services: TFTP ON
DNS/Web Server	192.168.100.20	VLAN 100	Services: DNS, HTTP ON
Mgmt-PC	192.168.50.10	VLAN 50	Admin workstation

— End of Document —

Router & Switch Management with TFTP Backup and IOS Upgrade
Cisco Packet Tracer 9.0 | University Network Project